

# Spanish → Chanka (Ayacucho) Quechua Machine Translation

From a 40.55 baseline to a  $\approx 46.7$  ChrF result — a conservative engineering report

rosettia · AmericasNLP 2021 spa→quy benchmark · single reference, ChrF (word\_order=0)

**Abstract.** We build a machine-translation system for Spanish → Chanka/Ayacucho Quechua (quy), an extremely low-resource, agglutinative language. By stacking **orthogonal** levers — a curated Qwen-9B LoRA chain, an NLLB-1.3B supervised model with synthetic distillation, character-level Minimum-Bayes-Risk decoding, cross-architecture ensembling, and GSPO reinforcement learning (to our knowledge the **first** application of GSPO to an encoder-decoder NMT model) — we improve from **40.55** to  $\approx 46.7$  ChrF on the AmericasNLP 2021 test, roughly **+7** over the best published system. We report everything conservatively: a single benchmark, single-reference ChrF, and **no native-speaker evaluation**. We also document, in detail, what did **not** work.

## 1 The problem

Chanka (Ayacucho) Quechua is Southern Quechua: agglutinative/polysynthetic (long, regular suffix chains), with non-standardized orthography. The resource situation is the binding constraint of the whole project:

- **Parallel data:** 100–300k noisy Spanish–Quechua pairs (Bible/JW300-derived, dictionaries, FLORES-200, mined).
- **Monolingual quy:** 175k lines.
- **Benchmark:** AmericasNLP 2021 spa→quy test, 1003 sentences, **single reference**.
- **Metric:** ChrF (sacrebleu, word\_order=0). Character-level metrics are the right choice for agglutinative morphology; BLEU/TER are unreliable here (see §6).
- **Hard constraint:** no Chanka speakers were available to us, so there is **no human evaluation** anywhere in this work. Every “quality” signal beyond ChrF is an automatic proxy.

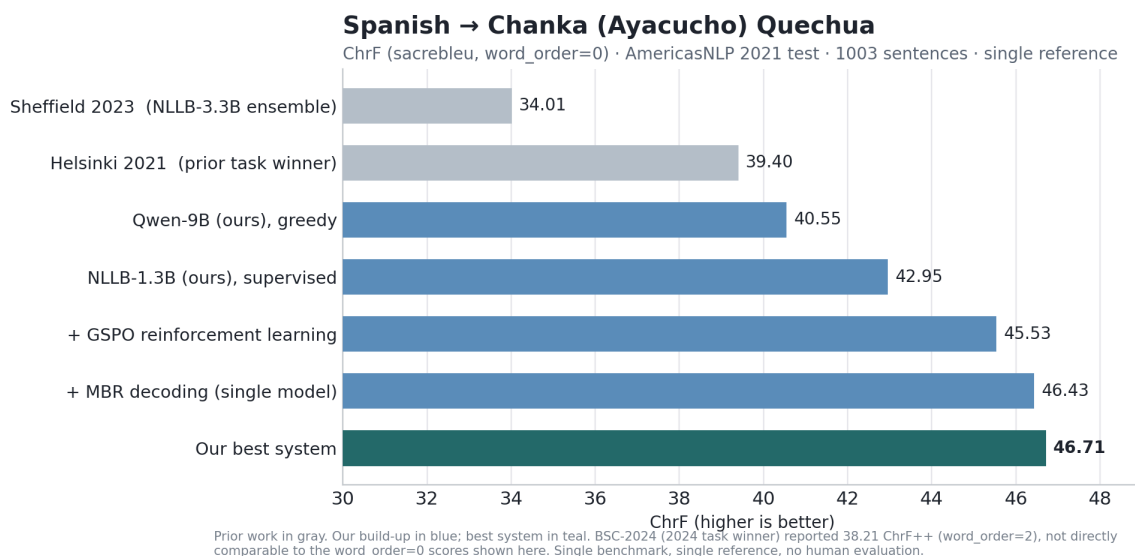


Figure 1: Our build-up against prior published systems. All scores are ChrF (word\_order=0).

## 2 Prior work (same test)

| System                                                     | ChrF (w0)        |
|------------------------------------------------------------|------------------|
| Sheffield 2023 — NLLB-3.3B, 3-model ensemble               | 34.01            |
| Helsinki 2021 — prior task winner                          | 39.40            |
| BSC 2024 — 2024 task winner (reported <b>ChrF++</b> 38.21) | ≈ low/mid-30s w0 |

Table 1: Published ceiling on this test sits 34–39.4 ChrF (w0). Note BSC-2024 reported ChrF+ + (word\_order=2), which reads 2–3 points above w0 and is not directly comparable.

## 3 What worked — the levers

### 3.1 Qwen-9B supervised model (v30)

Our base/teacher model, **v30**, is a **two-stage** LoRA fine-tune of unsloth/Qwen3.5-9B (Unsloth, **16-bit LoRA — no 4-bit quantization**, bf16; decoder-only, with minimal native Quechua pretraining), then merged to 16-bit weights. Both stages use LoRA **r = 256,  $\alpha$  = 512, dropout = 0** over all seven attention/MLP projections (q,k,v,o,gate,up,down), optimizer adamw\_8bit, weight-decay 0.01, warmup-ratio 0.05:

| Stage                     | Data                                  | LR   | eff. batch | max-seq | budget          |
|---------------------------|---------------------------------------|------|------------|---------|-----------------|
| Broad SFT                 | 166k spa↔quy (SomosNLP + AmericasNLP) | 1e-4 | 64 (16×4)  | 256     | → ckpt-2688     |
| Chanka SFT (continuation) | 1,929 curated Chanka pairs            | 2e-5 | 8          | 128     | 3 ep (ckpt-615) |

Table 2: The v30 two-stage recipe. The Chanka stage continues the broad adapter, then the result (v30a, ckpt-615) is merged to 16-bit (no  $\alpha$ -rescaling) → the published model.

The Chanka corpus (1,929 unique pairs after leakage-filtering against the held-out eval split) is 1,042 reviewed judicial-manual pairs + 503 manual-glossary entries + 349 Benito-2018 dictionary entries + 35 simple terms. Prompting uses a chat template (system **“Eres un traductor profesional español-quechua”**, instruction **“Traduce del español al quechua chanka...”**), with reasoning disabled and loss computed on the response only. **Result: 40.55 ChrF (w0) greedy** (already above the published ceiling), 42.93 with ChrF-MBR dedup decoding.

Two honest notes. **(i)** A compact-mixed self-verification stage and a LoRA- $\alpha$  sweep (carried over from a smaller-model lineage) were **planned but NOT included in the shipped/evaluated v30** — the SOTA number is the two-stage merge above. **(ii)** Curated small data beat larger noisy data — a 109k normalized in-domain corpus scored only 37.9, below the 1,929 curated pairs — so for a decoder-only model with little Quechua pretraining, curation and decoding mattered more than scale. (The in-domain manual eval is inflated because the glossary shares the manual the eval is drawn from; the AmericasNLP test is the clean, leakage-guarded number.) v30 is reused as the synthetic-data teacher and as an ensemble member; its full model card is linked at the end.

### 3.2 NLLB-1.3B supervised (BSC-2024 recipe)

LoRA r256/ $\alpha$ 512, lr 2e-4 inverse-sqrt. Standalone **39.46** — a strong, architecturally diverse second model (encoder-decoder, with real Quechua pretraining inside NLLB-200).

### 3.3 Synthetic distillation (first breakthrough, +3.5)

We forward-translated 200k Spanish monolingual sentences with the Qwen-9B teacher (greedy), filtered by length ratio, and added them as target-side synthetic data → retrained NLLB (“NLLB-r2”): **39.46 → 42.95**. This is sequence-level knowledge distillation; the gain is

bounded by the teacher ( 40-level), so the student approaches but does not surpass it from self-distillation alone.

### 3.4 Character-level MBR decoding

Reference-free Minimum-Bayes-Risk with a ChrF utility (COMET is unusable — its encoder has no Quechua). Two findings that matter: **(i)** deduplicate the candidate pool before taking the consensus (+0.5 over raw MBR — duplicates bias the centroid toward high-probability-but-not-best modes); **(ii)** suppressing the apostrophe token at decode is a small free gain (Ayacucho quy has no glottalization). Candidate-pool **temperature  $\approx 0.7$**  matters: MBR is sensitive to pool diversity.

### 3.5 Cross-architecture ensemble

Pooling candidates from architecturally diverse, **comparable-quality** models and taking the dedup-MBR consensus reached **45.01** (Qwen-9B @ NLLB-r2). The empirical rule: diversity helps only at comparable quality — a weak member (a regressed Qwen at 34.5, a zero-shot MADLAD at 19.78) **lowers** the ensemble.

### 3.6 GSPO reinforcement learning (headline, novel)

To our knowledge the **first application of GSPO** (Group Sequence Policy Optimization, Zheng et al. 2025) to an encoder-decoder NMT model. Setup: policy = NLLB-r2 LoRA; frozen NLLB-r2 as the KL reference; **reward = sentence-ChrF vs unseen Ayacucho references**; sequence-level (length-normalized) importance ratio + group-relative advantage + KL, with single inner-epoch updates. Rollouts run through an **in-process vLLM engine** — we implemented NLLB/M2M-100 support for vLLM (unsupported upstream) and sync policy weights into the engine each step (GPU→GPU). Reward = ChrF was selected on validation over ChrF++, length-penalty, repetition-penalty, and round-trip-adequacy variants.

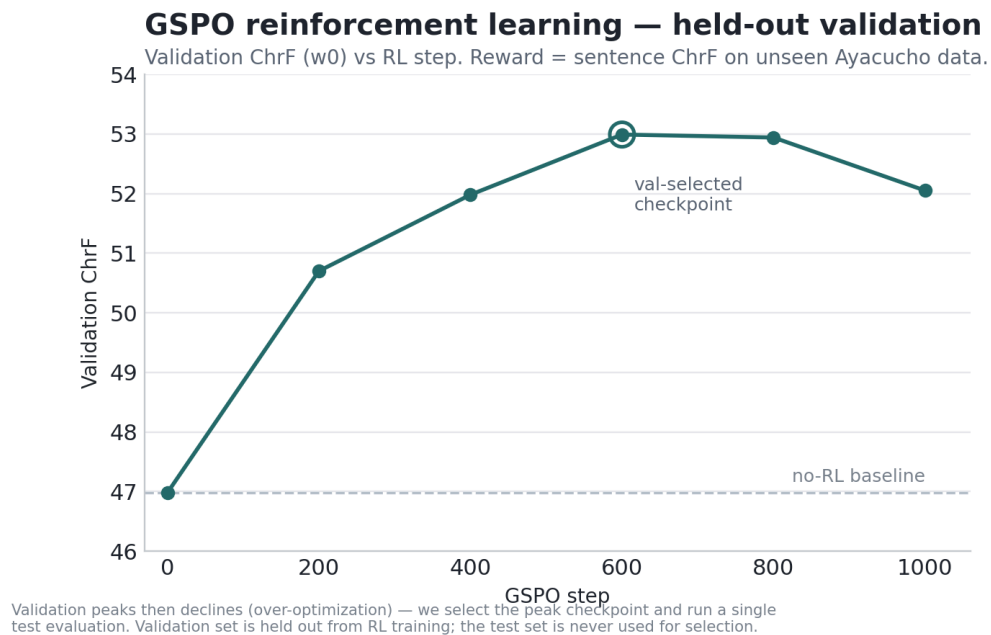


Figure 2: GSPO validation curve. Validation ChrF climbs, peaks ( step 600), then declines (over-optimization). We select the peak checkpoint on validation and evaluate on test once.

| System                                                    | ChrF (w0)    |
|-----------------------------------------------------------|--------------|
| Qwen-9B (ours), greedy                                    | 40.55        |
| NLLB-1.3B (ours), supervised + synthetic                  | 42.95        |
| Qwen-9B $\oplus$ NLLB-1.3B ensemble (ours)                | 45.01        |
| 1. GSPO RL, single model, beam5 ( <i>pre-registered</i> ) | <b>45.53</b> |
| 1. MBR decoding, single model                             | 46.43        |
| GSPO multi-checkpoint MBR ensemble (best-found)           | <b>46.71</b> |

Table 3: Progression on the AmericasNLP 2021 spa→quy test. See §8 on which number is the cleanest to cite.

## 4 Beyond the surface metric

ChrF is a single-reference surface metric and saturates. To check the RL gains are **real** quality and not metric-gaming, we score several automatic, speaker-free axes. GSPO improved **adequacy** (round-trip quy→spa vs the original Spanish source) by **+4.6** — more than it improved ChrF itself (+2.4) — and also reduced Spanish-word leakage and length error.

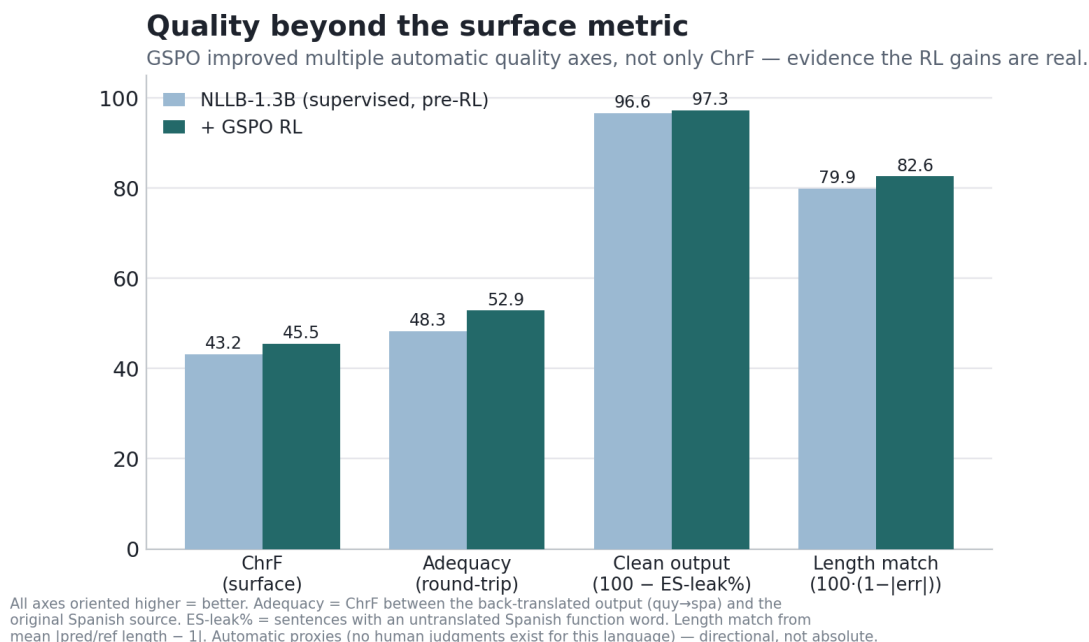


Figure 3: Multi-axis quality scorecard. All axes oriented higher = better. Automatic proxies — directional, not absolute.

## 5 Standard MT metrics

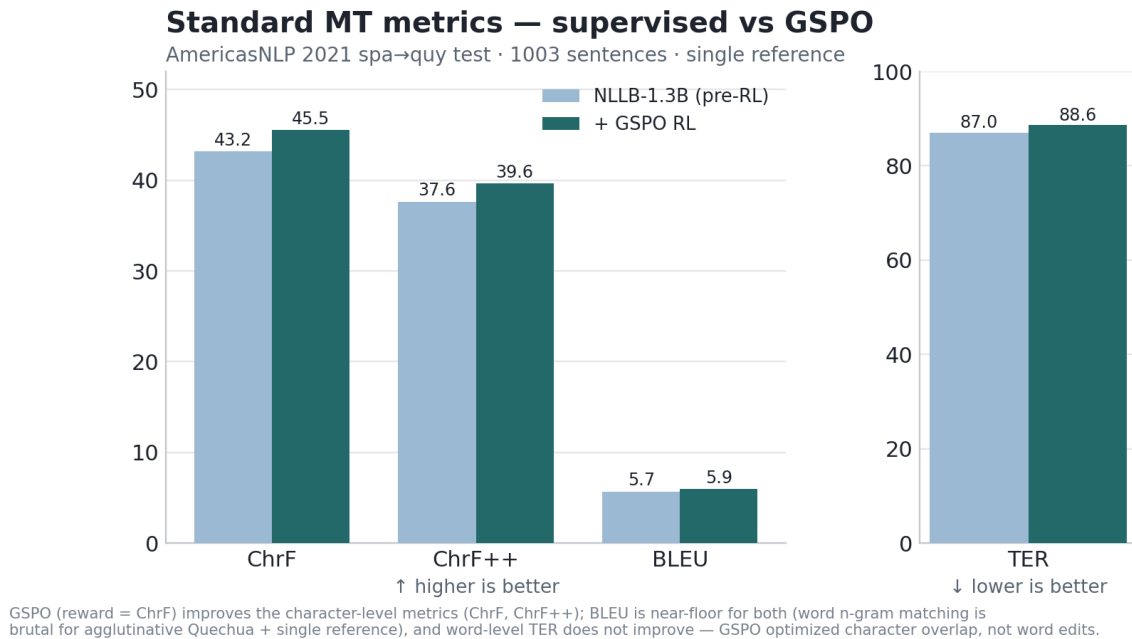


Figure 4: Standard metrics, supervised vs GSPO. GSPO (reward = ChrF) improves the character-level metrics (ChrF, ChrF++) and BLEU marginally; word-level **TER does not improve** — the RL optimized character overlap, not word edits. BLEU is near-floor for both (word n-gram matching is brutal for agglutinative Quechua under a single reference), which is why we treat ChrF as primary.

## 6 What did NOT work (honest)

We think the negative results are as informative as the positive ones.

- **MADLAD-400-3B (zero-shot):** wrong dialect (Cuzco-flavored), 19.78 ChrF — ruled out as an ensemble member; a weak member lowers the ensemble.
- **ByT5 (byte-level):** attractive in principle (char-level, orthogonal errors) but did not beat the subword NLLB here.
- **Round-trip / back-translation reward:** we built a reverse quy→spa model and a reward that scores adequacy by translating the candidate back to Spanish and comparing to the source. We **gated** it before committing a long run: its within-source ranking correlation with true quality plateaued at 0.25, and a controlled GSPO ablation confirmed it does **not** beat plain ChrF (val 49.63 vs 50.05). Falsified — cheaply, because we gated it.
- **A learned metric (COMET) for Chanka:** data-blocked. A deep, source-verified search found **no human-judgment MT dataset for any Quechua variety**; the only Indigenous MT-metrics task (AmericasNLP 2025) covers Guarani/Bribri/Nahuatl, not Quechua. Without human judgments (and with no speakers to create them) a human-grounded metric is not achievable; only proxies are.
- **HRM-Text-1B (a hierarchical-recurrent reasoning LM):** a curiosity experiment. From a cold start (English-only pretraining, English tokenizer that fragments Quechua 2×, no MT) it learned correct Chanka **morphology** surprisingly fast, but adequacy capped it in the mid-20s ChrF — far below NLLB, exactly as the tokenizer/no-pretraining ceiling predicts.
- **Engineering pitfalls:** a PEFT `.base_layer` weight-sync bug initially synced a half-base model into vLLM (reward stuck 22) until fixed; NLLB’s 256k vocabulary makes the log-prob forward memory-heavy (OOM unless micro-batched); and GSPO over-optimizes (validation peaks then declines), so checkpoint selection on a held-out validation split is essential.

## 7 Honesty and caveats on the numbers

- **Single benchmark, single-reference ChrF, no human evaluation.** ChrF  $\approx 46$  means roughly half the character n-grams match one reference — useful as a draft, not production quality.
- **Which number to cite.** The **pre-registered** result is the validation-selected checkpoint with our standard decode (beam5 + apostrophe-suppression): **45.53**, a single test evaluation. The MBR (46.43) and multi-checkpoint-ensemble (46.71) numbers involve decode-time choices — MBR temperature and which checkpoints to ensemble — that we **compared on the test set**; they are best-found configurations, and the 0.3–0.6 spread among them is within the noise of a 1003-sentence single-reference test. The robust, conservative claim is  **$\approx 46$  ChrF, clearly above prior published work.**
- **Dialect:** Ayacucho/Chanka (quy); not validated for Cuzco or Central varieties.

## 8 Reproduce

Scripts are in the repository (`scripts/`). High level: (1) `nllb/train_nllb_chanka.py` for the BSC-recipe SFT + synthetic distillation; (2) `rl/gspo_nllb_vllm.py` for GSPO (requires our NLLB-in-vLLM fork; reward = held-out-reference ChrF,  $G=16$ ); (3) select the peak checkpoint on the held-out validation split and run one test evaluation with `nllb/eval_nllb_americasnlp.py`; (4) optionally `decoding/gen_candidates_nllb.py` + `decoding/ensemble_mbr_rerank.py` for MBR/ensemble; and `decoding/quality_scorecard.py` for the beyond-the-metric audit.

## 9 Links

- Code: <https://github.com/Sekinal/rosettia-chanka>
- Model (LoRA adapter): <https://huggingface.co/Thermostatic/rosettia-quy-gspo-nllb13b-lora>
- Model (merged, standalone): <https://huggingface.co/Thermostatic/rosettia-quy-gspo-nllb13b-merged>
- Qwen-9B model (v30 — base/teacher/ensemble member, with its own training card): <https://huggingface.co/Thermostatic/rosettia-quy-v30b-9b-merged>
- NLLB/M2M-100 support for vLLM (our fork): <https://github.com/Sekinal/vllm/tree/add-nllb-m2m100-support>
- GSPO: Zheng et al. 2025, *Group Sequence Policy Optimization*, arXiv:2507.18071.